The Cox–Ingersoll–Ross Process: Moments, the Feller Condition, and Full Truncation

`hasklib` mathematical companion

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Abstract

Unlike GBM, OU, and ABM, the Cox–Ingersoll–Ross (CIR) process Cox, Ingersoll & Ross (1985) has no closed-form path solution derived in this note: neither the $\ln x$ substitution nor an integrating factor removes the x-dependence from its square-root diffusion coefficient. What the library actually needs from this process — its mean and variance, the parameter condition (Feller's condition) guaranteeing it stays strictly positive, and the motivation for the numerical scheme used when that condition is in doubt — is derived instead. The exact transition law (a scaled noncentral chi-squared distribution) exists but is not derived here; deriving it requires machinery (the squared-Bessel-process connection, or a PDE/Laplace-transform argument) well beyond this note's scope.

1 The CIR SDE

Definition 1 (Cox–Ingersoll–Ross process)**.** X_t follows a CIR process with mean-reversion speed $\kappa > 0$, long-run mean $\theta > 0$, and volatility $\sigma > 0$ if it solves

$$\mathrm{d}X_t = \kappa(\theta - X_t)\,\mathrm{d}t + \sigma\sqrt{X_t}\,\mathrm{d}W_t, \qquad X_0 > 0. \tag{1}$$

In the notation of `ito_foundations.tex` this is $a(t,x) = \kappa(\theta - x)$ (identical to OU) and $b(t,x) = \sigma\sqrt{x}$.

Remark 1 (Correspondence with the code)*.* `CIR::drift` returns $\kappa(\theta - x)$, exactly matching `OU::drift`. `CIR::diffusion` returns $\sigma\sqrt{x}$, which requires $x \geq 0$ to be well-defined — unlike GBM's σx or OU's constant σ, this diffusion coefficient is not defined for negative arguments at all, which is the source of every complication in this file.

Remark 2 (A genuine gap in the existence–uniqueness story)*.* The existence–uniqueness assumption in `ito_foundations.tex` requires a and b to be Lipschitz. $\sqrt{x}$ is not Lipschitz near $x = 0$ (its slope $\to \infty$ there), so that assumption does not literally cover (1) as stated. A strong solution nonetheless exists and is unique — this requires a separate, more delicate existence–uniqueness theory for Hölder-continuous (rather than Lipschitz) diffusion coefficients; see Cox et al. (1985) or Brigo & Mercurio (2006). *(Cited, not proved, in the same spirit as the original assumption — the library depends only on the conclusion.)*

2 Why the earlier tricks do not apply here

GBM's diffusion σx is proportional to the state, which $\ln x$ removed. OU's diffusion σ is constant, which the integrating factor $e^{\kappa t}$ exploited alongside an affine drift. CIR's drift $\kappa(\theta - x)$ is the same affine form as OU's, so the integrating factor idea is tempting to reuse — but it was the diffusion term, not the drift, that made OU's trick work cleanly: applying Itô's lemma to

$f(t,x) = e^{\kappa t}x$ needs $f_{xx} = 0$ to kill the second-order term entirely, which held because f was linear in x. The moment we try to fold $\sigma\sqrt{x}$ into any single substitution $f(t,x)$, the resulting f_{xx} term no longer vanishes, and no elementary choice of f linearises (1) the way $\ln x$ or $e^{\kappa t}x$ did for the earlier processes. This is not a failure of effort — $\sqrt{x}$ is a genuinely different kind of nonlinearity (a square-root, or CEV-type, diffusion), and no algebraic trick of this kind exists for it in general.

Remark 3 (The exact law exists, but is out of scope here). CIR's transition density is nonetheless known exactly: conditional on X_t, the (scaled) law of $X_{t+\Delta t}$ is a noncentral chi-squared distribution, a result following from CIR's connection to squared Bessel processes (Cox et al. 1985). This is the route `CIR::sample_terminal` would take for an exact sampler (Boost.Math's noncentral chi-squared machinery, per the library's `stats/` layer), analogous to `GBM::sample_terminal` and `OU::sample_terminal`. Deriving that transition density, however, requires either the squared-Bessel-process identification or a PDE/Laplace-transform argument neither established nor needed elsewhere in this note; it is cited, not derived, here.

3 Moments, via an ODE rather than a path solution

Although no closed-form X_t is derived, its first two moments are obtainable directly: take expectations of the SDE (or of Itô's lemma applied to a power of X_t) and solve the resulting deterministic ODE. This sidesteps needing a path solution at all.

Proposition 1 (Mean of CIR). *Let $m(t) = \mathbb{E}[X_t \mid X_0]$. Then $m(t)$ solves $m'(t) = \kappa(\theta - m(t))$ with $m(0) = X_0$, giving*

$$m(t) = X_0\, e^{-\kappa t} + \theta\big(1 - e^{-\kappa t}\big),$$

identical in form to OU's mean (`ou.tex`).

Proof. Taking expectations of the integral form of (1),

$$m(t) = X_0 + \kappa\theta \int_0^t \mathrm{d}s - \kappa \int_0^t m(s)\,\mathrm{d}s + \mathbb{E}\left[\int_0^t \sigma\sqrt{X_s}\,\mathrm{d}W_s\right].$$

The stochastic integral has zero expectation (the adapted Itô isometry cited in `schemes.tex`; Shreve (2004), Øksendal (2003)), regardless of the specific form of the diffusion coefficient inside it. Differentiating the remaining deterministic relation gives $m'(t) = \kappa\theta - \kappa m(t)$, exactly the deterministic ODE behind OU's mean, with the same solution. □

Remark 4. The mean depends only on the drift, and CIR's drift is literally the same function as OU's; the two processes' means therefore coincide exactly, even though their diffusion coefficients — and hence their variances, paths, and positivity properties — are entirely different.

Proposition 2 (Variance of CIR). *Let $v(t) = \mathrm{Var}(X_t \mid X_0)$. Then*

$$v(t) = \frac{\sigma^2 X_0}{\kappa}\Big(e^{-\kappa t} - e^{-2\kappa t}\Big) + \frac{\sigma^2\theta}{2\kappa}\Big(1 - e^{-\kappa t}\Big)^2.$$

Proof. Apply Itô's lemma to $f(x) = x^2$: $f_x = 2x$, $f_{xx} = 2$, $f_t = 0$. With $a = \kappa(\theta - X_t)$ and $b = \sigma\sqrt{X_t}$,

$$\mathrm{d}(X_t^2) = \Big(2X_t\cdot\kappa(\theta - X_t) + \tfrac{1}{2}\cdot\sigma^2 X_t\cdot 2\Big)\mathrm{d}t + 2X_t\cdot\sigma\sqrt{X_t}\,\mathrm{d}W_t = \big(2\kappa\theta X_t - 2\kappa X_t^2 + \sigma^2 X_t\big)\mathrm{d}t + 2\sigma X_t^{3/2}\,\mathrm{d}W_t.$$

Taking expectations and writing $w(t) = \mathbb{E}[X_t^2]$, the stochastic term again vanishes in expectation, giving

$$w'(t) = \big(2\kappa\theta + \sigma^2\big)m(t) - 2\kappa\, w(t).$$

Rather than solve for $w(t)$ directly, substitute $w(t) = v(t) + m(t)^2$ and use $m'(t) = \kappa(\theta - m(t))$ from Proposition 1 to simplify. Differentiating $w = v + m^2$ gives $w' = v' + 2mm' = v' + 2\kappa\theta m - 2\kappa m^2$. Substituting into the equation for $w'(t)$ and cancelling the $2\kappa\theta\, m(t)$ terms on both sides,

$$v'(t) + 2\kappa\theta\, m(t) - 2\kappa\, m(t)^2 + 2\kappa\big(v(t) + m(t)^2\big) = \big(2\kappa\theta + \sigma^2\big)m(t) \implies v'(t) + 2\kappa\, v(t) = \sigma^2 m(t),$$

with $v(0) = 0$ (since X_0 is a fixed starting point, not itself random). This is a much cleaner ODE than the one for $w(t)$: the θ-dependence has been absorbed entirely into $m(t)$, which is already known. Solve by the same integrating-factor technique used for OU's SDE, now applied to this ordinary (non-stochastic) ODE: multiplying by $e^{2\kappa t}$ collapses the left side into $\frac{\mathrm{d}}{\mathrm{d}t}\big(e^{2\kappa t}v(t)\big)$, and substituting $m(t)$ from Proposition 1 and integrating from 0 to t (using $v(0) = 0$) gives, after simplification,

$$v(t) = \frac{\sigma^2 X_0}{\kappa}\big(e^{-\kappa t} - e^{-2\kappa t}\big) + \frac{\sigma^2\theta}{2\kappa}\big(1 - e^{-\kappa t}\big)^2,$$

which is the claimed formula. *(The intermediate integrating-factor algebra is routine and omitted; it is the same mechanical steps as the OU variance derivation in `ou.tex`, applied to this ODE instead of directly to the SDE.)* □

Remark 5 (What this moment computation buys, and what it does not). Propositions 1 and 2 give exact first and second moments with no approximation — useful for moment-matching tests analogous to `test_stochastic.cpp`'s GBM check. They do not give the full distribution of X_t: CIR is not Gaussian (unlike OU or ABM), since $\sqrt{X_t}$ scales the noise unevenly across the state space. Recovering the full law requires the noncentral chi-squared transition density flagged above, not this ODE method.

4 The Feller condition

Theorem 1 (Feller condition). *If $2\kappa\theta \geq \sigma^2$, then $X_t > 0$ for all $t \geq 0$, almost surely, given $X_0 > 0$. If $2\kappa\theta < \sigma^2$, X_t can reach 0 in finite time (though it is instantaneously reflected back into $(0, \infty)$ by the drift, since $a(t, 0) = \kappa\theta > 0$).*

Citation. This is Feller's boundary classification for one-dimensional diffusions, applied to (1); see Cox et al. (1985) or Brigo & Mercurio (2006). *(Cited, not proved — the proof requires the scale-function/speed-measure machinery for one-dimensional diffusions, which nothing else in this note depends on.)* □

Remark 6 (Why this condition matters here specifically). Every other assumption in this companion (Lipschitz coefficients, $\kappa > 0$ for OU's stationary law) has been a technical convenience. This one is load-bearing in a different way: (1) is not even well-defined pathwise if X_t goes negative, since $\sqrt{X_t}$ has no real value there. The Feller condition is what guarantees the SDE never has to confront that question in continuous time. Nothing in `hasklib` currently enforces $2\kappa\theta \geq \sigma^2$ on constructing a `CIR` instance; whether to validate it, and what to do if a calibrated parameter set violates it (a common occurrence in practice), is a live design question, not a settled one.

Remark 7 (Relevance beyond the pricing showcase). CIR is, alongside Hull–White, one of the classical short-rate models in Brigo & Mercurio (2006) — valued for keeping rates strictly positive when the Feller condition holds, in contrast to Hull–White's Gaussian (and hence potentially negative) short rate. The parameter tension is real: market-calibrated CIR parameters frequently fail the Feller condition, which is precisely why the numerical scheme discussed next is not a mere implementation detail.

5 Motivation for full truncation

Even when the Feller condition holds and the true process X_t never reaches zero, a naive Euler–Maruyama discretisation (`schemes.tex`) can still fail: the update

$$\hat{X}_{t+\Delta t} = \hat{X}_t + \kappa(\theta - \hat{X}_t)\,\Delta t + \sigma\sqrt{\hat{X}_t}\,\sqrt{\Delta t}\,Z$$

has no mechanism preventing $\hat{X}_{t+\Delta t} < 0$ for an unlucky draw of Z — and once that happens, $\sqrt{\hat{X}_{t+\Delta t}}$ at the next step is not a real number at all. This is a discretisation artefact: it can occur even when the Feller condition guarantees the true continuous-time process stays positive, simply because a discrete-time approximation is not required to inherit that guarantee.

Definition 2 (Full truncation scheme)**.** Write $x^+ = \max(x, 0)$. The full truncation Euler scheme Lord, Koekkoek & van Dijk (2010) replaces every appearance of $\hat{X}_t$ inside the drift and diffusion evaluations — but not the running value being updated itself — by its positive part:

$$\hat{X}_{t+\Delta t} = \hat{X}_t + \kappa\big(\theta - \hat{X}_t^+\big)\,\Delta t + \sigma\sqrt{\hat{X}_t^+}\,\sqrt{\Delta t}\,Z.$$

Remark 8 (Why both terms are truncated, not just the diffusion). An earlier, more obvious fix (partial truncation) replaces $\hat{X}_t$ by $\hat{X}_t^+$ only inside the square root, leaving the drift's mean-reversion term $\kappa(\theta - \hat{X}_t)$ using the raw (possibly negative) value. Lord et al. (2010) found this still carries a systematic bias, and their contribution was to show that truncating the drift's evaluation as well — Definition 2 — reduces that bias further, at no extra implementation cost. The running value $\hat{X}_t$ itself is still allowed to go negative and stay negative for a stretch (it is only clipped where it is used, inside the coefficient evaluations); this is a deliberate feature of the scheme, not an oversight, and is what distinguishes "full" from "partial" truncation.

Remark 9 (Correspondence with the code). This is the discretisation `hasklib`'s roadmap specifies for CIR (OU → ABM → CIR with full truncation → Milstein), in preference to plain `EulerMaruyama::step` applied naively: wherever `CIR::drift`/`CIR::diffusion` are evaluated inside a stepping scheme, the state passed in should be $\hat{X}_t^+$, not $\hat{X}_t$ directly. Establishing the strong-order convergence properties of this scheme is a separate question from motivating it and is not addressed here.

References

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